

TinyML Smart Plug for Complementary Protection Against Progressive Fire-Inducing Electrical Faults in Residential Circuits

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Abstract—Miniature circuit breakers remain essential but mainly address major overcurrent and short-circuit events, leaving progressive socket-level fire-inducing conditions less directly observed. This paper presents a TinyML Smart Plug for complementary socket-level protection against sustained overload, degraded socket-contact heating, and load-limited series arc faults in 230V residential circuits. The prototype integrated calibrated sensing, ESP32-S3 control, relay-based load isolation, Firebase logging, and Progressive Web App (PWA) monitoring and labeling. Local firmware handled thermal estimation, threshold protection, load-context classification, Random Forest series-arc inference, fault latching, and relay actuation. Evaluation included sensor validation, controlled overload, socket-heating, and induced series-arc tests. Revalidation produced mean absolute errors of 3.910V, 0.178A, and 1.02°C. The highest-stress overload case reached 17.30A mean current and disconnected after 61.3s. Degraded sockets heated faster than normal sockets: at about 9A, relay trip occurred at 52min for the corroded socket and 120min for the normal socket; at about 13A, the corroded and normal sockets tripped at 5min and 46min, respectively. Across 200 balanced normal and arc trials, the embedded model and firmware persistence logic achieved 97.50% accuracy, 98.97% precision, 96.00% recall, and 97.46% F1-score. The results support controlled laboratory feasibility for a smart plug that complements mandatory residential electrical protection.

Keywords—ESP32-S3, series arc detection, socket-contact heating, sustained overload, Random Forest, load-type classifier, multi-domain sensing.

I. INTRODUCTION

Electrical faults remain a persistent residential fire concern. Philippine fire incident reporting identifies electrical issues as a leading cause of fires, while broader home-fire statistics also associate electrical failure or malfunction with structure fires, injuries, and property losses [1], [2]. The problem is not limited to catastrophic short circuits. It also includes progressive electrical hazards that may remain energized while producing thermal or waveform abnormalities before conventional circuit protection is forced to operate.

Conventional residential protection remains essential. Branch-circuit breakers, leakage protection, surge protection, and certified wiring practice address the fault classes for which they are designed. However, ordinary overcurrent protection is coordinated mainly around conductor thermal protection and high-current events; it does not directly detect progressive contact heating at a socket, persistent sub-trip overload behavior,

or low-current series arc faults that occur below the breaker immediate trip region [3]–[5]. At the plug-socket interface, a poor or degraded connection can generate localized heating under otherwise ordinary current flow. A load-limited series arc can also disturb the current waveform without producing a large RMS-current increase.

This study distinguishes three progressive fire-inducing fault conditions. Sustained overload is a current-persistence and time-stress problem. Degraded socket-contact heating is a localized thermal problem in which the hottest contact region may be hidden from an external thermistor. Series arc faults are stochastic and load-dependent because normal nonlinear loads can produce waveform behavior that may resemble arc-like disturbances. For this reason, the prototype does not treat all abnormal conditions as one generic fault state. Each condition requires protection basis that matches its physical behavior.

The objective of this work was to design, develop, and evaluate an ESP32-S3-based TinyML Smart Plug that provides complementary socket-level protection for selected progressive fire-inducing electrical faults in 230V residential circuits. The work integrated calibrated voltage, current, and temperature sensing; threshold-based overload and socket-heating protection; load-context classification; Random Forest series arc-fault inference; relay-based load isolation; fault latching; local audio-visual alerts; Firebase logging; and Progressive Web App (PWA) monitoring, event review, and data labeling.

The prototype uses a hybrid protection approach because the three target fault conditions do not produce the same protection basis. Sustained overload is handled through RMS-current level and persistence; degraded socket-contact heating is handled through device temperature measurement and compensated estimated socket temperature; and series arc-fault detection is handled through load context, waveform features, Random Forest inference, and firmware persistence logic. The evaluation was conducted under controlled laboratory conditions involving overload, induced series arc fault, and socket/contact-heating scenarios. The device is treated as a complementary protective layer that operates alongside, not in place of, circuit breakers, AFCIs, GFCIs, wiring inspection, and code-compliant residential protection.

II. DESIGN BASIS AND SYSTEM OVERVIEW

A. Target Fault Classes

The system was designed around progressive socket-level hazards. Overload was treated as a sustained load-current condition. A warning may be useful when the current is high, but immediate relay operation for every short transient or inrush event would conflict with time-current protection behavior. Thus, overload requires current persistence.

Degraded socket-contact heating was treated separately because poor contact pressure, corrosion, looseness, or oxidation can increase local contact resistance. Glowing-contact and receptacle-fire studies show that this kind of poor connection can persist while the circuit still carries load current [6], [7]. Recent poor-contact fire studies likewise show the role of contact pressure, localized heating, pyrolysis, and ignition progression [8], [9]. Consequently, a local thermal estimate is needed even when current alone cannot resolve the socket-contact condition.

Series arc faults were treated as a waveform-inference problem. A series arc is load-limited, so it may not create a breaker-level current rise. Instead, it can produce low-current gaps, current fluctuation, zero-region behavior, harmonic disturbance, and high-frequency content [5], [16]–[19]. Because normal loads can also be nonlinear, the arc decision benefits from load context and family-specific waveform interpretation [20], [21].

The design basis is summarized in Table I.

TABLE I
FAULT TYPES AND PROTECTION BASIS

Fault Type	Observed Behavior	Protection Basis
Sustained overload	Thermal stress from persistent load current	RMS-current threshold plus approximately 60s persistence score
Socket heating	Localized poor-contact heating at plug-socket interface	Thermistor input plus compensated estimated socket temperature
Series arc	Load-limited intermittent conduction and waveform disturbance	Load context, waveform features, Random Forest arc inference, and persistence counter

Each protection path asks a different question: whether current is persistently high, whether the socket interface is thermally abnormal, or whether the current waveform is repeatedly consistent with arcing under the present load context.

B. Hybrid Architecture

Fig. 1 shows the system architecture used in the prototype. The sensing layer measures line voltage, load current, and temperature near the plug/socket interface. The firmware layer performs calibration, filtering, RMS calculations, temperature estimation, waveform-feature extraction, load-context classification, TinyML arc inference, protection-state evaluation, relay control, and event logging. The cloud/PWA layer provides visualization, history review, labeling, and operator interaction. The relay decision remains local to the firmware.



Fig. 1. System architecture of the TinyML Smart Plug.

The architecture is intentionally hybrid. Deterministic thresholds are used where the protection evidence is direct and interpretable, as in overload current and estimated socket temperature. TinyML is used where the evidence is multi-feature, load-dependent, and ambiguous, as in series arc-fault detection. The resulting protection process is local: even if Wi-Fi or Firebase is unavailable, the ESP32-S3 can still evaluate faults and engage relay isolation.

III. METHODOLOGY

A. Hardware and Sensing Architecture

Fig. 2 shows the block diagram of the hardware implementation. The prototype was built as a plug-through socket-level device for a 230V residential circuit. Its major elements were an ESP32-S3 microcontroller, ZMPT101B voltage-sensing channel, ACS37035 current sensor, MCP3204 external 12-bit ADC, second-order anti-aliasing circuit, 10kΩ NTC thermistor, solid-state relay, relay latch, OLED display, buzzer, low-voltage supply, and enclosed outlet assembly.

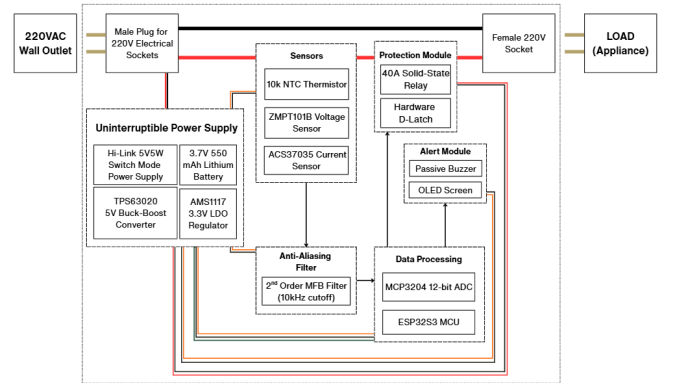


Fig. 2. Block diagram of the prototype hardware.

The current channel was central to the design. It supplied RMS current for overload protection, a current-dependent term for socket-temperature estimation, and waveform samples for feature extraction. The conditioned current waveform was digitized through the MCP3204 and processed by the ESP32-S3. The voltage channel supported RMS-voltage supervision and mains-present interpretation. The thermistor was placed near the plug-side wire-contact region and insulated to reduce direct ambient-air influence. Because this point is necessarily offset from the hidden contact hotspot, the thermal protection was based on compensated estimated socket temperature, which represents the available external thermal measurement rather than a direct internal-contact measurement.

The relay with a D-latch was the protective actuator. After a confirmed trip condition, the relay disconnected the protected

outlet and the fault state remained latched until cleared. The OLED, buzzer, and PWA provided alerts and user visibility. Firebase carried live telemetry, history, CSV review, and operator fields, whereas the relay actuation path was implemented in firmware and hardware.

B. Firmware Protection Flow

Fig. 3 summarizes the firmware protection flow implemented in the prototype. On startup, the ESP32-S3 initializes the sensors, ADC, relay driver, latch state, OLED, buzzer, Wi-Fi/Firebase interface, and protection manager. The main loop then acquires slow supervision variables and waveform frames in parallel: RMS voltage, RMS current, thermistor temperature, and current-frame features.

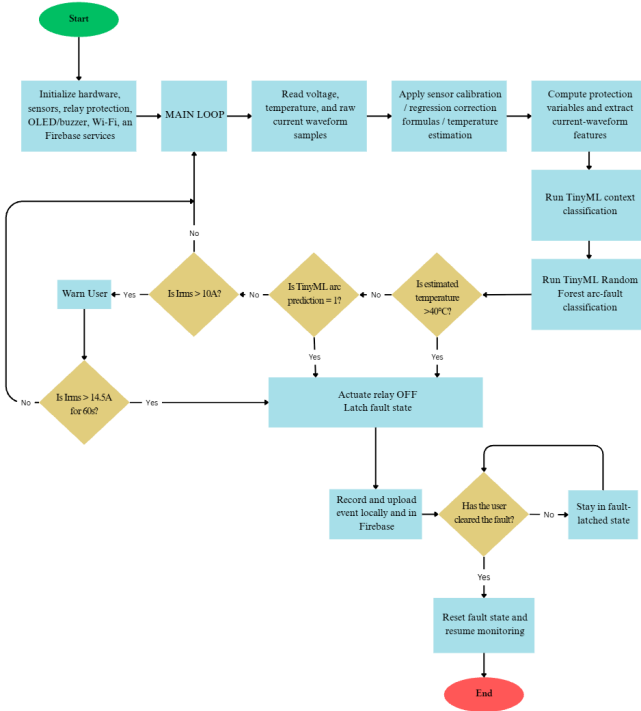


Fig. 3. Firmware flow for fault detection, relay actuation, and reset.

The final relay decision occurs in the protection manager. Calibrated voltage and current values feed deterministic voltage and overload states. The temperature routine produces device temperature and estimated socket temperature for thermal warning and trip evaluation. The waveform path computes feature frames, updates load context, runs Random Forest arc inference, and passes the arc output to a leaky persistence counter. If overload, socket heating, or series arc evidence reaches its trip condition, the firmware pulses the relay latch, records the fault, and reports the fault state locally and through Firebase when available. After relay isolation, the plug remains in a latched fault state until the user clears the fault through local or PWA-supported interaction.

Socket heating had highest reported-fault priority because temperature is the most direct local fire-risk indicator. Arc

TABLE II
FIRMWARE PROTECTION CONSTANTS

Function	Warning condition	or model	Relay-trip condition
Voltage supervision	Outside band	200-250V	Instant 170/260V, or 7s delayed low/high voltage
Overload	$I_{rms} > 10.0A$ warning		1s average current $\geq 14.5A$ for about 60s
Socket heating	39.0°C estimated temperature		Estimated socket temperature $\geq 40.0^\circ C$
Series arc fault	Context-aware Random Forest output		Arc counter reaches 8; +2 for arc frame, -6 for normal frame

fault was evaluated next because a series arc may be hazardous without sustained overcurrent. Overload remained a relay-trip condition after the persistence criterion was satisfied. The protection constants are listed in Table II.

C. Sensor Calibration and Validation

Voltage and current calibration used polynomial correction because the raw sensor outputs showed offset and scale nonlinearity across the tested range. The voltage channel was calibrated and revalidated against a Fluke 115 True RMS multimeter. The current channel was calibrated using appliance load combinations and revalidated against an Ingco clamp meter. The thermistor channel used the NTC divider and Beta-equation conversion, then was compared against an infrared thermometer at the external thermistor measurement point.

The validation procedure assessed protection-grade suitability rather than traceable metrology. Voltage validation checked whether calibrated readings were adequate for supply-state supervision. Current validation checked whether RMS-current error was small relative to warning and trip thresholds. Temperature validation checked the thermistor measurement at its own location only. Hidden-contact temperature was treated separately through degraded-contact experiments and conservative thermal estimation, because standards and thermal-measurement literature caution that external temperatures may lag or under-read internal contact points [10]–[14].

D. Overload Protection Method

Overload testing used high-power load combinations connected through the prototype. The firmware issued warning-level alerts above 10.0A while retaining relay closure during warning-level excursions. A sustained trip required the 1s current average to remain at or above 14.5A long enough for the overload score to reach six. Since the score increments every 10s, this corresponds to approximately 60s of persistent overload. This design choice prevents momentary current peaks and short transients from being treated as confirmed overload trips while still responding to persistent current stress.

The overload score can be represented by

$$S_{OL}[k] = \begin{cases} \min(6, S_{OL}[k-1] + 1), & I_{avg} \geq 14.5A, \\ \max(0, S_{OL}[k-1] - 1), & I_{avg} \leq 14.1A. \end{cases} \quad (1)$$

The relay trips when $S_{OL} = 6$. The method was evaluated by recording mean voltage, mean current, peak current, warning state, relay state, and trip latency during each load case.

E. Socket-Heating Method

Socket-heating evaluation used normal and intentionally degraded socket contacts under matched load cases. The degraded-contact trials served as accelerated laboratory comparisons rather than direct household field-fault replicas, allowing controlled observation of contact-heating behavior. For each trial, the prototype recorded device temperature from the external thermistor, estimated socket temperature from firmware, and relay state. An infrared thermometer supplied a reference terminal or contact-region reading for interpretation.

The firmware thermal model separated three thermal quantities. First, T_{NTC} was the direct external thermistor or device temperature. Second, $T_{normal,exp}$ was the expected normal socket temperature for a healthy contact carrying the measured current history. Third, $T_{socket,est}$ was the relay-facing estimated socket temperature used by the protection manager. This separation is needed because the thermistor is outside the hidden contact hotspot. It may lag or under-read the terminal region, while poor contact can generate localized heating that RMS current alone cannot represent.

The estimator was implemented as a compact firmware model. The current-shape term was computed as

$$s_I = \left[\text{clamp} \left(\frac{I_{rms}}{12.0}, 0, 1.25 \right) \right]^2, \quad (2)$$

so the thermal term follows the current-squared tendency of resistive heating while limiting extreme extrapolation. The expected normal term used a slowly filtered healthy-contact rise, whose active-load target was $\Delta T_{normal,target} = 0.20 + 12.0s_I$, added to the tracked ambient or baseline temperature T_a :

$$T_{normal,exp} = \text{clamp}(T_a + \Delta T_{normal}, -20, 125). \quad (3)$$

For an active load, the measured-temperature compensation was

$$\Delta T_{meas} = 0.30 + 1.20s_I, \quad (4)$$

with the current-dependent rise forced toward zero when the load was inactive. The protection-facing estimate was

$$\begin{aligned} T_{comp} &= \text{clamp}(T_{NTC} + \Delta T_{meas}, -20, 125), \\ T_{socket,est} &= \text{clamp}(\max(T_{comp}, T_{normal,exp}), -20, 125). \end{aligned} \quad (5)$$

The $\max(\cdot)$ operation prevents the relay-facing estimate from falling below the expected normal socket heating for the same load while still allowing measured local heating to dominate. The trip rule was applied to $T_{socket,est}$ so that raw thermistor readings and current-dependent normal heating both influenced the thermal state. The estimate should be interpreted as a conservative relay-facing variable, with its behavior checked against normal and corroded-socket experiments under comparable loads.

F. TinyML and Arc-Fault Method

Series arc detection used a two-stage load-aware TinyML pipeline. The method followed the firmware order: current-frame acquisition, waveform feature extraction, load-context classification, arc inference, and persistence-filtered relay evaluation. The firmware computed 512-sample current frames at a 27kHz target sampling rate with a 256-sample hop. A Hann-windowed frequency analysis and time-domain calculations generated waveform features; the Hann window was used to reduce spectral leakage in discrete Fourier analysis [25]. Logged features supported training, PWA review, and manual labeling.

The resulting TinyML processing sequence is shown in Fig. 4.

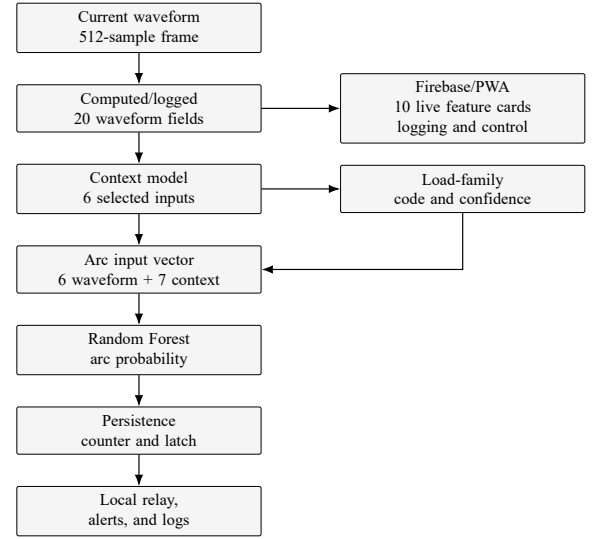


Fig. 4. TinyML processing flow for load-aware arc detection and relay actuation.

In this sequence, a centroid-based context model classified load family using six features: absolute RMS-current change, half-cycle asymmetry, midband residual ratio, current THD, mid/high-frequency spectral flux, and high-frequency energy delta. The firmware context schema defined six possible family fields: resistive linear, inductive motor, rectifier SMPS, phase-angle controlled, brush/universal motor, and other mixed. The collected and tested signatures in this study were limited to resistive loads, inductive loads, and brushed or universal-motor loads. The context model standardized its six inputs, computed distance to the stored load-family centroids, and accepted a family only when the best confidence was at least 0.45. Otherwise, the arc model used its stricter unknown-context threshold. This step was included because the same waveform evidence has different meaning under different loads: harmonic content and high-frequency variation may be normal for a brushed motor but unusual for a resistive heater.

Only features used by the context model or by the deployed arc model are listed in Tables III and IV. The notation follows the firmware feature frame: $i_c[n]$ is the cleaned current sample,

$N = 512$, f_s is the sampling rate, M_k is FFT magnitude, P_h is harmonic tone power, and ϵ is a small numerical offset. Symbols beginning with B denote adaptive healthy baselines maintained by the firmware.

TABLE III
CONTEXT-MODEL FEATURE COMPUTATIONS

Feature	Firmware computation	Context information
ΔI_{rms}	$ I_{\text{rms}}[k] - I_{\text{rms}}[k-1] $	Load step and operating-state change
Half-cycle asymmetry	$100 I_{\text{pos}} - I_{\text{neg}} /[0.5(I_{\text{pos}} + I_{\text{neg}})]$	Positive/negative cycle imbalance
Current THD	$100\sqrt{\sum_{h=2}^{10} P_h/P_1}$	Harmonic load signature
HF energy delta	$10\log_{10}[(r_{\text{HF}} + \epsilon)/(B_{\text{HF}} + \epsilon)]$	High-frequency content relative to baseline
Midband residual ratio	$20\log_{10}(R_{\text{mid}}/B'_I)$	Non-harmonic residual structure
Mid/HF spectral flux	$50\sum_{k=k_0}^{k_1} \hat{M}_k - \hat{M}_{k,\text{prev}} $	Frame-to-frame spectral change

TABLE IV
ARC-MODEL WAVEFORM FEATURE COMPUTATIONS

Feature	Firmware computation	Arc evidence represented
Max low-current run	$1000N_{\text{run,max}}/f_s$	Longest extinction or weak-conduction gap
Zero-dwell ratio	$100N_{\text{near zero}}/N_{\text{zc window}}$	Near-zero dwelling around crossings
Low-current ratio	$100N^{-1}\sum \mathbf{1}(i_c[n] \leq I_{\text{low}})$	Portion of weak-conduction samples
RMS-current z-score	$ I_{\text{rms}} - \mu_I /(\sigma_I + \epsilon)$	Load-relative current departure
Midband residual ratio	$20\log_{10}(R_{\text{mid}}/B'_I)$	Non-harmonic residual energy
Mid/HF spectral flux	$50\sum_{k=k_0}^{k_1} \hat{M}_k - \hat{M}_{k,\text{prev}} $	Nonstationary spectral disturbance

In Tables III and IV, $\hat{M}_k = M_k/\sum_{r=k_0}^{k_1} M_r$ is the normalized magnitude spectrum within the mid/high-frequency band, $r_{\text{HF}} = E_{\text{HF}}/(E_{\text{HF}} + E_{\text{UM}})$, and $R_{\text{mid}} = \sqrt{E_{\text{mid,res}}/N}$; $B'_I = \max(B_I, 0.10)$ limits division by very small baselines. The low-current threshold is $I_{\text{low}} = \max(0.035, 0.06\max(I_{\text{rms}}, 0.10))$, which keeps the conduction-gap tests load-relative while preserving a minimum near-zero threshold.

The arc model was a Random Forest deployed as a C header in firmware. Random Forest was selected because it can represent nonlinear feature interactions while remaining exportable as a compact embedded ensemble [23], [24]. The deployed model used 13 inputs: six waveform features and seven context fields. The waveform features were maximum low-current run, zero-dwell ratio, low-current ratio, absolute RMS-current z-score from baseline, midband residual ratio, and mid/high-frequency spectral flux. The context fields were

six one-hot load-family inputs and one context-confidence input. The arc probability threshold was 0.48 for supported confident contexts and 0.78 for unknown context, with an additional minimum-feature-vote safeguard for unknown loads. The selected Random Forest used 160 trees, a maximum depth of 16, minimum leaf size of 4, minimum split size of 12, bootstrap sampling, entropy splitting, and cost-complexity pruning alpha of 0.0001. The deployed feature set can be summarized in four groups. First, the conduction-gap inputs which are maximum low-current run, zero-dwell ratio, and low-current ratio, represent extinction, weak conduction, and near-zero dwell that can occur during separation and restrike. Second, the current-departure input, expressed as an absolute RMS-current z-score from the recent baseline, measures load-relative departure from healthy operation. Third, the waveform-disturbance inputs, namely the midband residual ratio and mid/high-frequency spectral flux, represent residual spectral energy and frame-to-frame spectral change associated with nonstationary arcing. Finally, the load-context inputs, composed of six one-hot load-family fields and the context confidence, condition the arc decision so that normal nonlinear loads are evaluated with context-sensitive decision boundaries.

Training and testing were performed offline from firmware CSV logs. Records were reviewed, manually labeled, cleaned with Python scripts, split into training, validation, and reserved test views, and exported as firmware headers. Fig. 5 summarizes this workflow from laboratory CSV records to embedded deployment. The final arc report used 35,512 training rows, 9,434 validation rows, and 10,268 reserved test rows. Embedded testing then evaluated the exported model inside the firmware protection path, where the model output was filtered by the arc persistence counter before relay action. Thus, the reported trial result represents the deployed protection pipeline, including embedded inference and counter-based relay logic. This is different from reporting an offline classifier alone: the relay result includes sensing, feature extraction, context gating, model output, persistence scoring, and latch actuation.

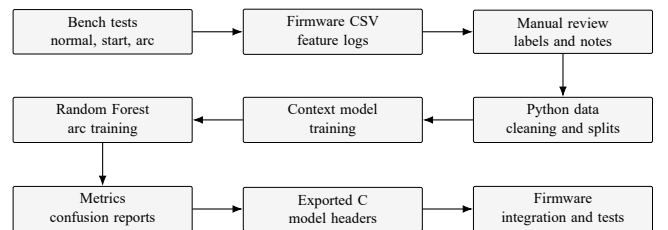


Fig. 5. Data labeling, model export, and embedded deployment workflow.

Table V gives the corresponding dataset-preparation counts. The small number of arc-label rows relative to normal-label rows explains why the embedded detector was evaluated using precision, recall, F1-score, and trial behavior rather than accuracy alone. It also explains why firmware persistence was used: a single arc-like frame was not treated as enough evidence for relay isolation.

TABLE V
ARC-FAULT DATASET PREPARATION SUMMARY

Preparation item	Count
Raw CSV rows before cleaning	55,085
Rows retained after cleaning	49,296
Rows removed during cleaning	5,789
Trainable rows	45,010
Normal-label rows	49,013
Arc-label rows	283
Arc-training rows	30,821
Load-context rows	18,475

G. PWA, Firebase, and Logging Role

The PWA and Firebase backend supported the research workflow by showing live voltage, current, device temperature, estimated socket temperature, context family, arc output, relay status, event history, and waveform features. They also supported CSV inspection, data labeling, and operator fault clearing. These functions improved observability and reproducibility during laboratory trials. The local ESP32-S3 firmware remained responsible for protection-state evaluation and relay isolation.

IV. RESULTS AND DISCUSSION

A. Sensor Validation

Table VI summarizes the revalidation results. Voltage MAE was 3.910V, current MAE was 0.178A, and temperature MAE was 1.02°C. These errors were small relative to the implemented thresholds. Current error, for example, was much smaller than the separation between the 10.0A warning threshold and 14.5A sustained-trip threshold. The temperature result validates the external thermistor measurement point only; the socket-heating protection result is interpreted through the compensated estimate and degraded-contact tests.

TABLE VI
SENSOR VALIDATION SUMMARY

Channel	MAE	MSE	RMSE
Voltage	3.910V	15.539V ²	3.94V
Current	0.178A	0.04076A ²	0.202A
Temperature	1.02°C	1.142°C ²	1.07°C

B. Overload Protection

Table VII gives the overload evaluation. All heavy-load trials exceeded the warning threshold, while relay operation occurred in the highest-stress case. Trial 4 averaged 17.30A, reached a 19.73A peak, and opened the relay after 61.3s. That trip time agrees with the intended approximately 60s persistence behavior. Trials 1, 3, and 5 reached peaks near or above the sustained threshold, then fell below the criterion before accumulating the required persistence score. Thus, the overload function behaved as time-aware complementary protection. Breaker coordination and formal time-current compliance testing require a separate standards-oriented evaluation.

TABLE VII
OVERLOAD EVALUATION RESULTS

Trial	Load setup (WH: Water Heater)	Mean (A)	Peak (A)	Relay trip
1	1.3kW kettle + WH	12.87	14.85	No
2	1.3kW kettle + 1.5kW kettle	12.33	13.63	No
3	1.5kW kettle + WH	13.36	14.86	No
4	1.3kW kettle + WH + 2 lamps	17.30	19.73	61.3s
5	1.3kW kettle + WH + 1 lamp	14.42	16.23	No

C. Socket-Heating Protection

Socket-heating characterization showed that contact condition changed the thermal outcome at similar current levels. The low-current protection case is shown in Fig. 6, while the severe high-current case is shown in Fig. 7. The paired figures show the intended relay behavior across the tested range: lower-current degraded-contact heating remained below the trip criterion, while the severe case reached the relay-facing thermal threshold.

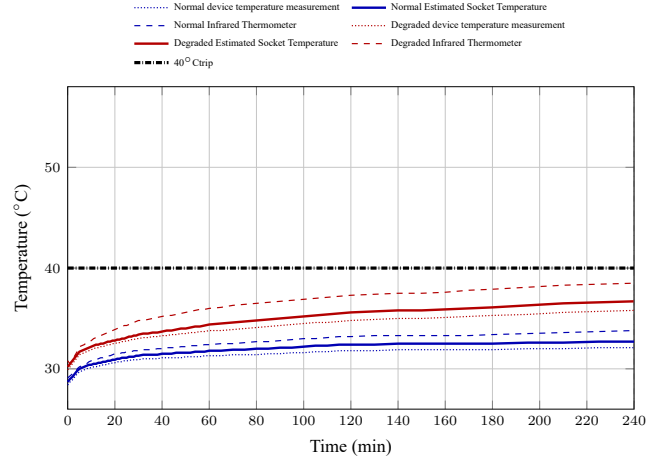


Fig. 6. Water-heater-only low-current socket-heating protection validation.

Table VIII summarizes all socket-heating protection trials. At about 5A in the water-heater-only case, neither socket tripped: the corroded terminal ended hotter than the normal terminal (38.5°C versus 33.8°C), but the estimated socket temperature remained below the 40.0°C relay criterion. At about 9A, the corroded socket reached the 38.2°C comparison point in 48min, compared with 105min for the normal socket. At about 11A, the corroded socket reached the same point in 7min, compared with 52min for the normal socket. In the severe 13A corroded case, the relay tripped after 5min and the reference terminal reached 56.0°C. These results support the premise that socket heating cannot be inferred from RMS current alone. They also show that the thermal branch is a protection function, not a corrosion classifier: at higher current levels, normal and corroded sockets can both reach the conservative relay-facing threshold, while the degraded

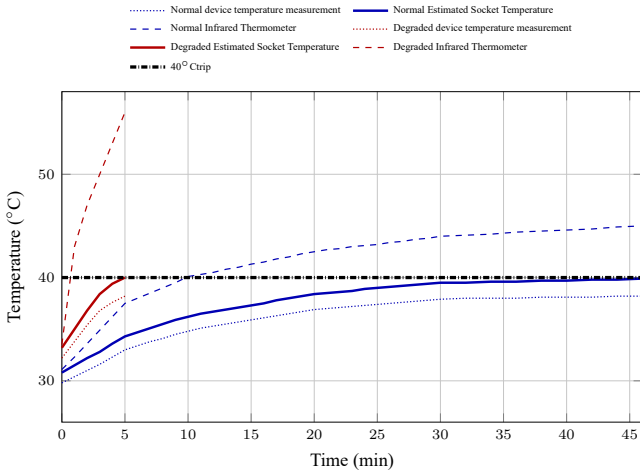


Fig. 7. Water heater plus four halogen lamps during high-current socket-heating protection validation.

socket shows faster rise and higher terminal temperature under comparable loading.

The difference between device temperature, estimated socket temperature, and IR terminal temperature is important. The prototype senses the contact hotspot indirectly through an external thermistor and a conservative relay-facing estimate. This approach provides socket-level action before contact heating progresses further while preserving non-trip behavior in low-current cases where the estimate remains below threshold. The comparison therefore supports a complementary trip strategy based on estimated thermal risk, while the IR terminal readings remain interpretive references rather than the quantity directly measured by the device.

TABLE VIII
SOCKET-HEATING PROTECTION SUMMARY

Load case (WH: Water Heater)	Socket	Mean (A)	IR terminal (°C)	Relay
WH	Normal	5.17	33.8	No
WH	Corroded	4.85	38.5	No
WH + 1 lamp	Normal	7.11	40.3	No
WH + 1 lamp	Corroded	7.10	42.4	Yes
WH + 2 lamps	Normal	9.06	43.2	Yes
WH + 2 lamps	Corroded	9.04	44.1	Yes
WH + 3 lamps	Normal	11.07	43.9	Yes
WH + 3 lamps	Corroded	11.14	55.0	Yes
WH + 4 lamps	Normal	13.02	45.0	Yes
WH + 4 lamps	Corroded	13.08	56.0	Yes

D. Series Arc and TinyML Results

The cleaned arc-training dataset contained 49,296 rows with 283 arc markers, confirming that arc evidence was sparse and imbalanced in the logged feature records, as shown earlier in Table V. This imbalance supports the design choice of interpreting relay-facing behavior at trial level, where embedded inference, context gating, persistence counting, and relay actuation are evaluated together.

The final relay-facing trial confusion matrix is shown in Table IX. Across 200 balanced normal and arc trials, the deployed model and firmware persistence logic detected 96 of 100 arc trials and produced one false trip in 100 normal trials. The corresponding performance metrics are shown in Table X. Overall trial-level accuracy was 97.50%, precision was 98.97%, recall was 96.00%, and F1-score was 97.46%. Trial-level metrics represent the complete firmware behavior: repeated arc-like outputs accumulate toward a relay decision, while isolated normal outputs decay.

TABLE IX
SERIES ARC-FAULT TRIAL CONFUSION MATRIX

Trial group	TP	FP	FN	TN
Recognized loads	49	0	1	50
Unrecognized loads	47	1	3	49
All tested loads	96	1	4	99

TABLE X
SERIES ARC-FAULT PERFORMANCE METRICS

Evaluation level	Accuracy	Precision	Recall	F1-score
Trial, recognized	99.00%	100.00%	98.00%	98.99%
Trial, unrecognized	96.00%	97.92%	94.00%	95.92%
Trial, all loads	97.50%	98.97%	96.00%	97.46%

TABLE XI
LOAD-FAMILY TRIAL RESULTS SUMMARY

Family	Tested loads	Arc detected	Normal correct
Resistive	Water heater, halogen lamp, electric kettle, heat gun	40/40	40/40
Inductive	40W stand fan	10/10	10/10
Brushed/universal	Handheld vacuum, Electrolux vacuum, angle grinder, bench cutter, impact drill	46/50	49/50

TABLE XII
PER-LOAD TRIAL-LEVEL ARC PERFORMANCE

Load	Family	Acc.	Prec.	Rec.	F1
Water heater	Resistive	100%	100%	100%	100%
Halogen lamp	Resistive	100%	100%	100%	100%
Electric kettle	Resistive	100%	100%	100%	100%
40W stand fan	Inductive	100%	100%	100%	100%
Handheld vacuum	Brushed	95%	100%	90%	94.74%
Electrolux vacuum	Brushed	95%	100%	90%	94.74%
Angle grinder	Brushed	95%	100%	90%	94.74%
Bench cutter	Brushed	95%	100%	90%	94.74%
Impact drill	Brushed	95%	90.91%	100%	95.24%
Heat gun	Resistive	100%	100%	100%	100%

The selected arc features were physically consistent with the recorded error pattern. Maximum low-current run, zero-dwell ratio, and low-current ratio responded to conduction interruption rather than to large RMS-current magnitude. The

RMS-current z-score expressed departure from the recent healthy baseline, so it remained load-relative. The midband residual ratio and mid/high-frequency spectral flux captured non-harmonic and nonstationary waveform disturbance after ordinary periodic structure was reduced. These quantities gave clear trial separation for the tested resistive and inductive loads, as shown in Tables XI and XII. The context features served a different role from the arc features. ΔI_{rms} and half-cycle asymmetry helped identify load state and waveform shape; THD, high-frequency energy delta, residual ratio, and spectral flux described the harmonic and nonstationary signature of the operating load. Appending the resulting family and confidence fields allowed the same arc-feature vector to be interpreted with different thresholds for supported and unknown contexts.

The remaining trial-level errors were concentrated in brushed or universal-motor loads. The impact drill contributed the single normal false trip, while several brushed-load cases contributed missed arc trials. This behavior is consistent with the feature interpretation: healthy commutation, startup, and nonlinear current in brushed motors can produce high harmonic content, spectral variation, and current irregularity that partially overlap with arc indicators. The context features in Table III were therefore important because they described the load family before applying the arc model; however, the tested context coverage remained limited to resistive, inductive, and brushed or universal-motor families. Broader household electronic-load coverage is required before the detector can be treated as generally representative of residential loads.

E. Integrated Protection Assessment

The integrated results show complementary behavior among the three protection paths. Overload protection addressed persistent current stress without tripping every transient. Socket-heating protection responded to degraded-contact thermal behavior that current alone could not identify. Series arc protection used waveform features and context to address a load-limited fault that may remain below conventional overcurrent operation. These results indicate the value of separating evidence by fire-inducing mechanism while combining the outputs in one local relay-protection pipeline.

The demonstrated behavior is bounded by the laboratory scope: selected loads, induced faults, and bench conditions. Certification, field readiness, and full residential nuisance-immunity evaluation require separate future work.

F. Scope and Limitations

The study was conducted under controlled laboratory conditions. Long-term deployment in occupied homes, aged installations, large appliance populations, full branch-circuit diversity, and naturally occurring faults were outside the evaluation scope. The load set and fault conditions were limited by safe bench testing, especially for induced series arcs.

The arc model may produce nuisance detections or missed detections on unseen loads. Although the firmware context schema retained six possible family fields, the collected and

tested load signatures were limited to resistive, inductive, and brushed or universal-motor loads. Sensitive SMPS loads, phase-angle-controlled loads, and mixed electronic loads were not included in comprehensive induced-fault coverage. The reported false positive and false negative cases show that the dataset remains incomplete, especially for brushed and non-linear loads. The reported trial-level metrics should therefore be read together with the confusion matrix and load-family summary, not as a universal nuisance-immunity claim.

The thermal sensing method is indirect. The thermistor was located outside the hidden contact spot, and the reference terminal reading often exceeded the external device temperature and estimated socket temperature. The 40.0°C threshold is a prototype control threshold for conservative relay action rather than a universal socket safety limit or direct hotspot measurement.

Wi-Fi and Firebase supported monitoring, logging, labeling, and operator interaction. Local relay protection was independent of cloud connectivity, although remote visualization and history features were reduced when the network was unavailable. The device provides supervision at one socket rather than whole-house fault localization.

Certification-oriented testing was outside the present work. Creepage, clearance, enclosure robustness, relay thermal rating, endurance, electromagnetic compatibility, abnormal operation, and regulatory safety tests remain future work. The implementation should be evaluated as complementary socket-level supervision alongside breakers, AFCIs, GFCIs, surge protection, code-compliant wiring, and professional electrical assessment.

V. CONCLUSION AND RECOMMENDATIONS

The TinyML Smart Plug demonstrated controlled laboratory feasibility as a hybrid socket-level protection implementation for selected progressive fire-inducing electrical faults. As an ECE implementation, it integrates calibrated sensing, deterministic protection logic, load-aware embedded TinyML, relay latching, and monitoring support in one local protection pipeline. The work also shows why the evidence must be separated by mechanism: overload requires current persistence, degraded socket-contact heating requires conservative thermal estimation, and series arcs require load-aware waveform inference.

Within the tested scope, sensor errors were acceptable for the implemented thresholds, the sustained overload function tripped the 17.30A case after 61.3s, degraded-contact tests showed faster and hotter socket behavior than normal contacts, and the deployed arc pipeline achieved 97.50% trial-level accuracy while retaining the four missed arc trials and one normal false trip in the analysis. These findings support feasibility for the proposed complementary protection architecture.

Future work should expand the normal, nuisance, and arc-fault dataset across more residential loads, plug conditions, contact gaps, and wiring arrangements. Longer semi-field testing should support later field-readiness evaluation. Thermal sensing should be improved through better placement,

multiple sensors, or stronger estimation models. Firmware and hardware should be optimized for longer sampling, logging reliability, and memory margin. Enclosure, insulation, relay rating, creepage, clearance, strain relief, and thermal safety should be redesigned toward certification-oriented testing. Arc labeling and trial annotation should also be improved so future TinyML models can be trained with stronger fault-transition evidence.

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